KAIZEN MATHS

Lesson starter

Number Bases and Number Sets

Base Conversion

Name _____

S1

Convert 11011 2 to denary.

S2

Convert 90 10 to base 5.

S3

In 1122 5 , find the value of the digit 1 in its position.

S4

Convert 1111 2 to denary.

Show your working

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NUM-5141-L2 · S

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Exit ticket

Number Bases and Number Sets

Base Conversion

Name _____

E1

Convert 137 10 to base 6.

E2

In 2222 5 , find the value of the digit 2 in its position.

E3

Convert 121 3 to denary.

KAIZEN MATHS

Exit ticket

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Base Conversion

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In 2222 5 , find the value of the digit 2 in its position.

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Convert 121 3 to denary.

NUM-5141-L2 · E

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NUM-5141-L2 · E

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Teacher answers & working

Number Bases and Number Sets · Base Conversion

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S1

Convert 11011 2 to denary.

Answer:

27_{10}

Working:

Use powers of the base as place values.

$11011_2 = 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

$1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 1 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1$

$1 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1 = 27$

So $11011_2 = 27_{10}$.

S2

Convert 90 10 to base 5.

Answer:

330_5

Working:

Divide repeatedly by 5 and record the remainders.

Divide by 5	Quotient	Remainder
$90 \div 5$	18	0
$18 \div 5$	3	3
$3 \div 5$	0	3

Read the remainders from bottom to top.

Therefore $90_{10} = 330_5$.

S3

In 1122 5 , find the value of the digit 1 in its position.

Answer:

25_{10}

Working:

The digit 1 is in the 5^2 place.

Value = 1×5^2

Value = $1 \times 25 = 25$.

Teacher answers & working

Number Bases and Number Sets · Base Conversion

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S4

Convert 1111 2 to denary.

Answer:

15_{10}

Working:

Use powers of the base as place values.

$1111_2 = 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

$1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 1 \times 8 + 1 \times 4 + 1 \times 2 + 1 \times 1$

$1 \times 8 + 1 \times 4 + 1 \times 2 + 1 \times 1 = 15$

So $1111_2 = 15_{10}$.

E1

Convert 137 10 to base 6.

Answer:

345_6

Working:

Divide repeatedly by 6 and record the remainders.

Divide by 6	Quotient	Remainder
$137 \div 6$	22	5
$22 \div 6$	3	4
$3 \div 6$	0	3

Read the remainders from bottom to top.

Therefore $137_{10} = 345_6$.

E2

In 2222 5 , find the value of the digit 2 in its position.

Answer:

50_{10}

Working:

The digit 2 is in the 5^2 place.

Value = 2×5^2

Value = $2 \times 25 = 50$.

Teacher answers & working

Number Bases and Number Sets · Base Conversion

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E3

Convert 121 3 to denary.

Answer:

16_{10}

Working:

Use powers of the base as place values.

$121_3 = 1 \times 3^2 + 2 \times 3^1 + 1 \times 3^0$

$1 \times 3^2 + 2 \times 3^1 + 1 \times 3^0 = 1 \times 9 + 2 \times 3 + 1 \times 1$

$1 \times 9 + 2 \times 3 + 1 \times 1 = 16$

So $121_3 = 16_{10}$.